

COMPUTER NETWORKS

PHASE 1 — LITERATURE REVIEW AND RESEARCH GAP IDENTIFICATION

The purpose of this phase is to develop the ability to critically read and analyze recent research literature related to your proposed research topic. You need to identify established methodologies, datasets, models, evaluation strategies, reported findings, limitations, and potential research gaps.

1. Paper Selection

Each group must identify and critically study three (3) Q1 journal research papers published in 2026 that are directly related to their proposed research topic.

PAPER SELECTION REQUIREMENTS

Each selected paper must:

- Be published in 2026.
- Be a research article, not merely an editorial, opinion article, book chapter, conference paper, news article, or general webpage.
- Be published in a Q1-ranked journal.
- Be directly relevant to the student's proposed research topic.
- Present an identifiable research methodology, experimental setup, or analytical framework.
- Include sufficient information about the dataset/data source, methods, and evaluation to allow critical analysis.
- Preferably be from a reputable publisher/journal such as Springer Nature, Elsevier, IEEE, Wiley, ACM, Taylor & Francis, or another recognized academic publisher.

REQUIRED DELIVERABLE:

ONE-PAGE CRITICAL SUMMARY THAT COVERS:

- Limitations explicitly mentioned by the authors.
- Limitations that you identify yourself.

- Dataset limitations.
- Methodological limitations.
- Evaluation limitations.
- Generalizability issues.
- Missing experiments.
- Potential directions for future research.

CROSS-PAPER COMPARISON TABLE:

#	Title	Year of Publication	Dataset Used	Preprocessing Techniques	Models / Techniques Used	Evaluation Metrics Used	Results	Limitations / Research Gap
1	Paper 1	2026						
2	Paper 2	2026						
3	Paper 3	2026						

IMPORTANT REQUIREMENT

The table should not simply copy sentences from the papers. Students must synthesize the information and make the differences between the three studies clear.

RESEARCH GAP SYNTHESIS:

Based on the analysis of the three papers, you need identify at least 3 potential research gaps.

Research Gap	Evidence from Paper 1	Evidence from Paper 2	Evidence from Paper 3	Proposed Research Opportunity
Gap 1				
Gap 2				
Gap 3				